Phase Equilibria

10/26/2018

1. (20 pts) What is the chemical potential difference of water and ice at -3.50 °C at 1 atm? $S^o_{H_2O,s} = 48.0 Jmol^{-1}K^{-1}$, $S^o_{H_2O,l} = 70.0 Jmol^{-1}K^{-1}$, assuming the S is independent of temperature.

2. (40 pts) The phase diagram of NH_3 can be characterized by the following information. The normal melting and boiling temperatures are 195.2 and 239.82 K, respectively; the triple point pressure and temperature are 6077 Pa and 195.41 K, respectively. The critical point parameters are 112.8×10^5 Pa and 405.5 K. Make a sketch of the P-T phase diagram (draw to scale whenever possible) for NH_3 and label the details according to the information provided above. Place a point in the phase diagram for the following conditions. State which and how many phases are present.
   a. 195.41 K, 1050 Pa
   b. 195.41 K, 6077 Pa
   c. 237.51 K, 101325 Pa
   d. 420 K, 130×10^5 Pa
   e. 190 K, 6077 Pa

3. (40 pts) Use the vapor pressures for C_2N_2 given in the following table to estimate
   (a) The enthalpies of fusion, vaporization, and sublimation. Assume that the enthalpies are independent of temperature.
   (b) The temperature and pressure of the triple point.

Phase	T (°C)	p (Torr)
Solid	-62.7	40.0
Solid	-51.8	100
Liquid	-33.0	400
Liquid	-21.0	760